

- **Docker and Kubernetes are popular tools in modern software development, used to build, deploy, and manage applications in a more efficient and scalable way.**
- **Docker is a platform that allows developers to create, deploy, and run applications inside containers. Containers are lightweight, isolated environments that package an application and its dependencies, enabling it to run consistently across different environments.**
- **Kubernetes (often abbreviated as K8s) is an open-source container orchestration platform that automates the deployment, scaling, and management of containerized applications.**



Hypervisor

Operating system

Hardware

Virtualized deployment



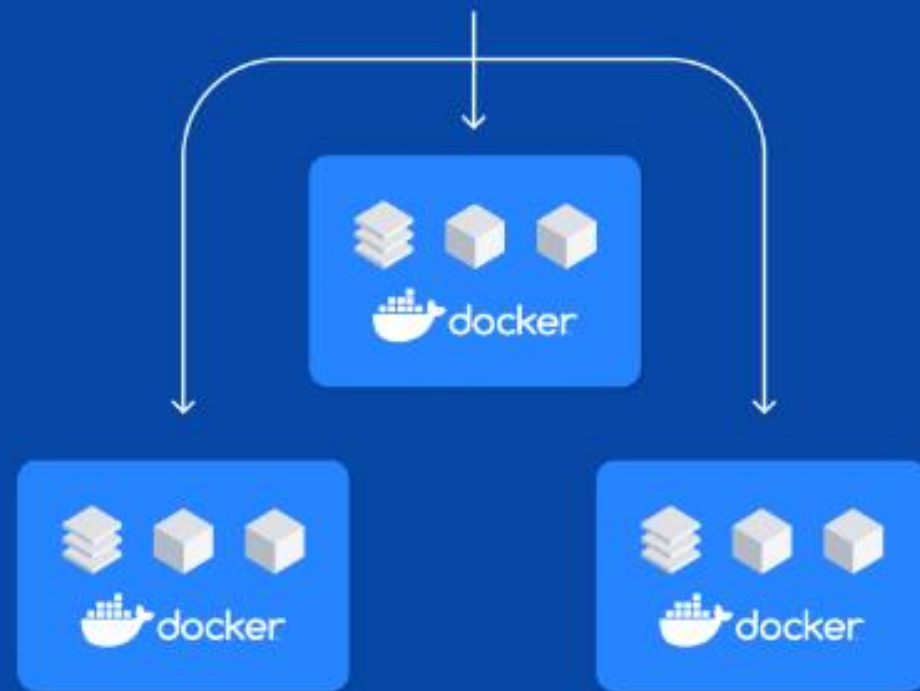
docker

Operating system

Hardware

Container deployment

 **kubernetes**



Kubernetes deployment

- **Docker creates images through a process that builds a set of instructions, written in a Dockerfile, into a layered, portable image.**
 - **Each layer represents an instruction in the Dockerfile, such as installing a package or copying files.**
- 1) Writing a Dockerfile: The Dockerfile is a plain text file containing instructions to set up the environment and define the application's behavior inside a container. Few instructions are FROM, RUN, COPY, CMD**
 - 2) Building the Image: With the Dockerfile ready, you can build an image using the docker build command. This command reads each instruction in the Dockerfile, executes it, and creates a new image layer.**

3) Layered Image Creation: Docker uses a layered file system to optimize storage and speed. Each command in the Dockerfile (like FROM, RUN, COPY) creates a separate layer in the image.

For example:

- Base layer (FROM python:3.9-slim): Downloads the base image.**
- Application code layer (COPY . /app): Copies files from the host machine.**
- Dependency installation layer (RUN pip install -r requirements.txt): Installs dependencies.**

4) Committing and Storing the Image

5) Running the Image